Metadata Creation Guidelines

This page outlines local content standards for **non-MARC descriptive metadata** created by Iowa State University Library for the description of digital objects and collections.

Metadata creation guidelines tell metadata creators how to input information into metadata elements or fields. These guidelines are known as content or input standards. They also may be called metadata best practices or style guides.

Content standards complement metadata schemas and application profiles (MAPs), which define the elements used in a particular repository or platform. Access MAPs created for Iowa State University Library repositories at Metadata Application Profiles and Crosswalks.

- For information about the different types of metadata standards, refer to Minitex's Standards for Digital Collections.
- For the MARC content standard, refer to Resource Description and Access (RDA).
- For the finding aid content standard, refer to Describing Archives: A Content Standard (DACS).
- For links to other style guides and standards, see the Resources section of this page.

Table of contents

- Metadata principles
- Goals of description
- General guidelines
  - Accuracy
  - Users' needs
  - Respect
  - Encoding
- Style guide
  - Capitalization
    - Sentence case
    - Names
    - Terms from controlled vocabularies
  - Spacing
    - Spaces
    - Line breaks
  - Spelling
    - Sources
    - Abbreviations
    - Misspellings
  - Punctuation

- Guidelines by element
  - Title element
    - Formal, devised, and remediated titles
    - Articles in titles
    - Capitalization in titles
    - Dates in titles
    - Names in titles
    - Publications in titles
    - Punctuation in titles
  - Name elements
  - Subject elements
    - Faceted approach
    - Assigning similar terms
    - Identity-related terms
    - Problematic and preferred subject terms
  - Date elements
  - Description element
    - Collection-level descriptions
    - Object-level descriptions
    - Alternatives to object-level descriptions
  - Extent element
  - Alt text
- Resources
- Sources

Metadata principles

Metadata Services, in collaboration with Digital Scholarship and Initiatives (DSI) and other library stakeholders, follows NISO's *Framework of Guidance for Building Good Digital Collections* (2007), which provides the following principles for metadata creation (pp. 61–62):

1. Good metadata conforms to community standards in a way that is appropriate to the materials in the collection, users of the collection, and current and potential future uses of the collection.
2. Good metadata supports interoperability.
3. Good metadata uses authority control and content standards to describe objects and collocate related objects.
4. Good metadata includes a clear statement of the conditions and terms of use for the digital object.
5. Good metadata supports the long-term curation and preservation of objects in collections.
6. Good metadata records are objects themselves and therefore should have the qualities of good objects, including authority, authenticity, archivability, persistence, and unique identification.

Goals of description

In creating descriptive metadata for digital objects and collections, Metadata Services works to meet the following goals:

1. Facilitate the discoverability and usability of digital objects by both people and systems.
2. Expose context for the original resource and the collection within which it is presented.
3. Enable users to assess the authenticity of the digital surrogate.
4. Provide access to materials in culturally appropriate and inclusive ways.
5. Demonstrate the library's commitment to access, scholarship, and reuse.
6. Illustrate the value of archives and cultural heritage materials to the library and university missions.

General guidelines

Accuracy

Record only factual information—that is, information found in the original resources, provided by creators, or verified in other authoritative sources. Do not rely on personal assumptions or inferences.

Users' needs

Consider how the metadata you create will help users to find, identify, select, obtain, and explore the resources you describe (*IFLA Library Reference Model*, "Users and User Tasks"). Will researchers use the terms in your description to search for and discover the resource? Will the metadata help them decide whether to access the resource? Will it provide context to enhance understanding of the resource?

Respect

Be aware that people depicted in or associated with described resources may see the metadata. Descriptions must be accurate, relevant, and respectful. When possible, use the terminology preferred by the people and communities you're describing. For living people especially, respect personal preferences and privacy.

Encoding

Encoding refers to the translation of characters into binary numbers for use by computers. UTF-8 encoding ensures that web pages can display Unicode characters. The Unicode character set includes far more diacritical marks, non-Latin characters, and symbols than the older ASCII character set.

- When generating CSV files, be sure they are encoded as UTF-8. If they are not, special characters may be rendered as ⁇.
- When reviewing data that may have been encoded incorrectly, you can find special (non-ASCII) characters with the regular expressions `[^\x20-\x7F]` (includes newlines) or `[^\x20-\x7F\n]` (excludes newlines).

> ⚠ UTF-8 encoding is the default when exporting CSVs from OpenRefine and Google Sheets. It is **not** the default encoding when exporting from Excel.

For UTF-8 and ASCII character codes, see W3Schools: https://www.w3schools.com/charsets/ref_html_utf8.asp; https://www.w3schools.com/charsets/ref_html_ascii.asp.

Style guide

> ⚠ These are general style guidelines. Rules may differ by the element and controlled vocabulary in use. For more information, consult the Guidelines by element section of this page and the relevant MAP (see Metadata Application Profiles and Crosswalks).

Capitalization

Sentence case

In general, use sentence case: capitalize proper nouns and the first word of each sentence or title.

> Photograph of people using the Dubuque municipal swimming pool

Names

- Follow the capitalization style used in the authorized form of the name, as shown in the Library of Congress Name Authority File (LCNAF) or Iowa State University Library Vocabularies (ISULV).
- If a name lacks an authorized form, follow the capitalization style used in the official website or other work created by the person or organization, the original resource, or other authoritative sources.

> hooks, bell, 1952-2021
>
> L'Oréal
>
> OpenRefine

Terms from controlled vocabularies

In subject, type, format, and genre elements, follow the capitalization style used by the selected controlled vocabulary.

For example, the Library of Congress Subject Headings (LCSH) and Faceted Application of Subject Terminology (FAST) vocabularies capitalize the first word of a subject heading, subdivision, or reference, while the Getty Art & Architecture Thesaurus (AAT) lowercases all terms that are not proper nouns.

LCSH subject term: Photography--Equipment and supplies

FAST subject term: Composition (Photography)

AAT genre term: fashion photographs

Spacing

Spaces

- Use a single space between sentences.
- In personal names, separate multiple initials with a single space.

S. R. Ranganathan

Line breaks

- In metadata creation sheets, avoid line breaks (newlines) within cells. These are created, for example, by pressing Alt+Enter.
- To find and replace newlines, use the regular expression `\n`.

Spelling

Sources

For correct spellings, consult the resource, the relevant controlled vocabulary, the Merriam-Webster dictionary, and other authoritative sources.

Abbreviations

- Spell out abbreviations that may not be understood.
- When an organization is known by its acronym or initialism, spell out the full name at first mention, followed by the acronym or initialism within parentheses.

Library of Congress Name Authority File (LCNAF)

Misspellings

- When transcribing a misspelling, add the correct spelling in square brackets after the misspelling.
- Do not use the Latin *sic*.

"... includes works by Christian Peterson [Petersen]"

Punctuation

The following table displays general rules regarding punctuation.

Punctuation mark	Rules	Examples
Ampersand	Avoid the ampersand (&) unless it is part of an authorized name.	Association for Information Science and Technology (ASIS&T)
Apostrophe	Use straight apostrophes ('), not curly ones (’). Style decades in numerals without apostrophes. For more guidance, see *The Chicago Manual of Style,* "Use of the apostrophe."	children's literature students' habits 1980s
Comma	Use the serial (Oxford) comma. In personal names, use commas to set off suffixes only when the name is inverted. For more guidance, see the comma-related sections of *The Chicago Manual of Style,* "Punctuation."	Correspondents included the president, the vice president, and the secretary of state. Martin Luther King Jr. King, Martin Luther, Jr., 1929-1968
Dash and hyphen	Use an em dash (—) or two hyphens (--) between phrases.	The specimens were small animals—for example, birds and rodents.

	Use a single hyphen (-) between digits and in compound words. For more guidance, see *The Chicago Manual of Style,* "Hyphenation guide."	Bird watching--Therapeutic use 1996-2001 two-thirds
Period	Include periods in abbreviations of personal and geographic names. In personal names, separate initials with a single space. Do not use periods in acronyms or initialisms. End descriptions and alt text with a period, even when they consist of a phrase rather than a complete sentence. In subject elements, do not add a terminal period to names or terms from controlled vocabularies.	W. E. B. Du Bois U.S. government NASA Describes the 1892 family trip to Italy and stops in Genoa, Rome, and Naples. Iowa State University. Special Collections and University Archives
Quotation mark	Use straight quotation marks ("..."), not curly ones (“...”). Place periods and commas within closing quotation marks. Use double quotation marks, not single quotation marks, to indicate quoted text. Use double quotation marks to set off *cited** titles of books, articles, songs, works of art, unpublished works, web pages, and so on. Impose sentence case.	"Let them eat cake" is a mistranslation of "Qu'ils mangent de la brioche." Burns wrote the article "Brighter futures for cats."

	*Cited titles typically appear in the description element. ⚠ In print, cited titles of books and some other works are italicized. However, as italics are not compatible with every online discovery platform, cataloging practice has been to use quotation marks for all cited titles. See *The Chicago Manual of Style*, "Italics versus quotation marks for titles."	

Guidelines by element

⚠ These are guidelines for the population of core metadata elements. Rules may differ by element.

ℹ Consult the relevant MAP (see Metadata Application Profiles and Crosswalks) for information about specific elements, including whether they are required or repeatable, their definitions and usage, the content standards and controlled vocabularies used to populate them, examples, and common crosswalks.

Title element

⚠ The guidelines below apply primarily to object or item-level titles. For guidelines on devising collection titles, refer to Digital Collection Title and Description Guidelines.

Formal, devised, and remediated titles

- If the resource has a formal title, transcribe it.
  - Impose sentence case.
  - Do not consider the name of a form or template printed on a document as a formal title.
- If the resource lacks a formal title, devise one.
  - Use sentence case.

- Include the nature of the resource and the topic.
- Add the date if needed for clarification or identification (see dates in titles below).
- For correspondence, record the name of the creator before the nature of the resource, followed by the name of the recipient.
- For more guidance, refer to DACS 2.3.

- If a formal title contains outdated or otherwise problematic language, consider remediating it, adding a content warning, or both.

Formal:

- Computing machine for the solution of large systems of linear algebraic equations
- Suffrage cook book

Devised:

- Dedication program for Dolliver Memorial State Park
- Nico Egan letter to Barack Obama

Remediated:

- Sculpture of an enslaved person in the gardens of Melbourne Hall *(formal title is "The Black Slave")*

Articles in titles

Avoid initial articles (a, an, the) for better alphabetization in search results.

Capitalization in titles

For transcribed and devised titles, use sentence case: capitalize the first word, proper nouns, and initialisms. Lowercase all other words.

Photograph of people using the Dubuque municipal swimming pool

Dates in titles

- For formal titles of serials, include volume, issue, and chronological information.
- For devised titles, include dates if they are needed for clarification or differentiation.
- Spell out months. (Use numeric dates *[YYYY-MM-DD]* only in date elements.)

AAAP spring newsletter, May 2005
Cora Mae Trice Greene letter to David Lendt, August 3, 1988

Names in titles

- Use the form of the name that appears in the resource.

- Enter the authorized form of the name in an agent or subject element.
- Include abbreviated honorifics—for example, Dr., Mrs., or Rev.—in transcribed titles but not devised titles.
  - Information about a person's honorific title may be added to the description element if necessary.

Publications in titles

- For articles and clippings, transcribe the formal title and append the title of the publication in parentheses.
- Impose sentence case.

If you're gay, wear denim (Iowa State daily)

Punctuation in titles

- Do not enclose devised titles in square brackets.
- Do not end titles with a period.
- Do not enclose titles in quotation marks.
  - **Exception:** Titles of works cited within an object title are enclosed in quotation marks.

Photograph of Amanda Gorman reciting "The hill we climb"

- Use a colon before a subtitle or subseries.
- Use a period before a part title (title of a volume or episode in a series).

Formal title with subtitle: Premonition: a pandemic story

Formal title with part title: Midwest farm report. Rural economic truth

Formal title with subseries title and part title: The Alaska earthquake, March 27, 1964: regional effects. Slide-induced waves, seiching, and ground fracturing at Kenai Lake

Name elements

Use the authorized or preferred form of personal and corporate names.

- In general, use the authorized form of a name, as shown in LCNAF.
- If the name does not exist in LCNAF, choose from a locally maintained vocabulary of names.
- If the name does not appear in a local vocabulary, use the spelling of the name as shown in the resource and follow LCNAF style: *[Last Name], [First Name], [birth year]-[death year]*
  - Do not include honorific titles such as "Dr." or "Mrs."
  - For women, do not include the husband's first name.

- For women, include a maiden name in name elements only when there is no established name authority **and** the maiden name either is needed for identification or is part of the preferred name.
  - Do not enclose a maiden name in parentheses; place it after the first and middle names.
  - When not including a maiden name in the name element, consider incorporating it into the description element.
- Prioritize preferred names when known. For verification, refer to the official website or other work created by the person or organization, the original resource, or other authoritative sources.
- If an LCNAF record or local vocabulary conflicts with a preferred name, consult with Metadata Services (metadata@iastate.edu) to determine next steps.

> Use the NACO board to make requests for name authority records to be created or revised through the Name Authority Cooperative Program (NACO).

Subject elements

Faceted approach

Assign individual subject terms rather than constructing complex terms with subdivisions. This approach:

- Collocates or groups items with similar broad themes.
- Facilitates faceted searching and filtering.

> For topical subject terms, Metadata Services recommends Faceted Application of Subject Terminology (FAST), a faceted vocabulary derived from the Library of Congress Subject Headings (LCSH).

Assigning similar terms

- You may assign related terms to the same resource if all terms are relevant—in other words, they all describe the "aboutness" of the resource.
- You may assign broader and narrower terms to the same resource to increase discovery.

Related terms: Veterinarians; Veterinary medicine
Broader and narrower terms: Black people; African Americans

Identity-related terms

Before assigning identity-related subject terms, review the following points:

- Do not infer race, ethnicity, gender, or other identities from a photograph without supporting evidence. For example, could people potentially described as African Americans be from outside the United States or not of African heritage?
- Consider whether information about identity is important for discovery and relevant to the topic of the resource. Irrelevant references to personal characteristics may "other" individuals, or imply they are outside the norm (see *The DEI Metadata Handbook,* "Inclusive description," and *MLA Handbook,* "Principles of inclusive language").
- Rather than assigning an identity-related term, Metadata Services recommends assigning multiple general terms (i.e., taking a faceted approach).

⚠ If you do choose to use an identity-related term, also assign a general term or terms. Otherwise, resources described with only identity-related terms may not be displayed with more general resources in search results or facets.

Identity-related term: Women scientists
General terms: Women; Scientists

Problematic and preferred subject terms

All controlled vocabularies may contain outdated, offensive, or otherwise problematic terms.

- In general, for maximum discovery, Metadata Services recommends retaining problematic terms and assigning preferred terms as supplemental subjects.

⚠ If a subject element is associated with a particular controlled vocabulary, such as FAST, you may need to place preferred terms in a separate element for keywords or local subjects.

- When retaining problematic terms, consider adding a content warning in a disclaimer or note element:

Iowa State University Library collects, preserves, and provides access to materials for teaching, learning, and research from a broad range of sources and time periods. Some of these materials may be offensive or contain negative stereotypes. Their presence should be viewed and considered within their historical context. They do not represent the views of Iowa State University or the University Library.

Date elements

In elements designated for dates only, use numeric notation. Follow the ISO 8601 standard or the related Library of Congress Extended Date Time Format (EDTF) Specification:

- Year: *YYYY*
- Year and month: *YYYY-MM*
- Full date: *YYYY-MM-DD*
- Range: *YYYY/YYYY*

1939
2003-09
1921-02-28
1950/1955

Description element

Collection-level descriptions

See Digital Collection Title and Description Guidelines.

Object-level descriptions

- Write one to three sentences about the resource.
- Begin the description with a characterization of the object ("Book," "Letter," "Sculpture," "Episode," etc.), followed by a brief explanation of the contents, form, or contextual history if relevant.
  - The first sentence may be a fragment and may begin without an article.
- End all descriptions with a period.
- Incorporate keywords: use words that make objects discoverable and usable by people and systems.
- Include relevant names and terms not used in the title, name, or subject elements.
- Use plain language: give preference to basic vocabulary and short sentence structure.
- Do not simply repeat the object title.
- Do not duplicate the alt text description.

Episode of the "Sew easy" television program produced by Iowa State College and sponsored by Home Craft Films. In this episode, host Lucille Rea discusses the unit method of construction, the layout of the sewing area, and getting your machine ready to sew.

- Enclose transcriptions of photograph captions or handwritten annotations in double quotation marks.

Annotation: "(left to right) Arlene Morris, James B. Morris III, William S. Morris"

- Do not use quotation marks for notes about the resource, such as content warnings or access restrictions.

Original manuscript available for viewing in the Special Collections and University Archives reading room at Iowa State University Library.

Alternatives to object-level descriptions

⚠ Descriptions are time consuming to write and to review for quality. If much of the descriptive information about a resource can be provided by the title, name, and subject elements, consider one of the following options.

- Do not populate the description element.
- Use the same description for each object in a collection or series by repurposing the higher-level description.

Item from an archival collection of news clippings and promotional materials about multicultural student organizations at Iowa State University.

Extent element

- Use a numeral to indicate the size or duration of the resource, followed by the appropriate extent unit—for example, an RDA Carrier Extent Unit.
- Do not capitalize the extent unit.
- For audiovisual materials, use the HH:MM:SS format. No extent unit is needed.

1 photograph
58 pages
00:24:58

Alt text

- Populate for non-textual objects so that screen readers can describe them.
- Briefly convey the information that is displayed visually, especially information not provided in accompanying metadata.
- Do not repeat the title or description, as screen readers will read out those elements.
- Do not begin with "Photograph of," as screen readers will announce images.
- End with a period so that screen readers will pause before moving on.
- See also *The DEI Metadata Handbook,* "Accessibility."

Aerial view of the library and surrounding park with trees in full bloom.

Resources

Chicago Manual of Style

DEI Metadata Handbook

Describing Archives: A Content Standard

Editorial Standards - Brand Standards

Home, Digital Collections - Iowa State University Library

Home, Iowa State University Library Vocabularies

Library Guides: Web Content Style Guide: Content Style Guide

Metadata Application Profiles and Crosswalks

Merriam-Webster: America's Most Trusted Dictionary

RDA Toolkit

Sources

Cataloging Cultural Objects (CCO)

CDP Dublin Core Metadata Best Practices Version 2.1

Metadata Application Profile

Creating Metadata

IFLA Library Reference Model: A Conceptual Model for Bibliographic Information

Metadata Entry Guidelines | Minnesota Digital Library

MODS Version 3 User Guidelines (Metadata Object Description Schema - MODS, Standards, Library of Congress)

Montana Memory Project: Metadata

NISO Framework of Guidance for Building Good Digital Collections

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